

TSM 098 | INDEPENDENT PRACTICE

Polynomial operations and basic factoring

While TSM 099 studies Topic 16. Review: Topic 9, Topic 10, Topic 11, Topic 12, Topic 14, Topic 15.

Name: _____ Date: _____

Complete the core pages first; use the extension page for remaining time. Show all work here or on separate paper. Check answers after attempting each section and correct errors.

Core: like terms and distribution

1. Combine like terms.

$$(x^2 + x - 1) + (-3x^2 - 2)$$

2. Combine like terms.

$$(-6x + 1) + (-2x^2 + 5x + 4)$$

3. Combine like terms.

$$(-3x^2 + 4x - 3) + (-2x^2 - 2x - 6)$$

4. Distribute and simplify.

$$2x(-x^2 - 3x + 3)$$

5. Distribute and simplify.

$$2x(-x^2 - 3x - 5)$$

6. Distribute and simplify.

$$2x(-x^2 - 3x - 4)$$

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Core: polynomial multiplication

7. Multiply and combine like terms.

$$(x - 5)(2x + 3)$$

8. Multiply and combine like terms.

$$(x + 4)(x + 1)$$

9. Multiply and combine like terms.

$$(x - 1)(2x + 1)$$

10. Multiply and combine like terms.

$$(x + 2)(2x - 4)$$

11. Multiply and combine like terms.

$$(x^2 + x + 1)(x^2 - x + 3)$$

12. Multiply and combine like terms.

$$(x^2 + x + 2)(x^2 - x + 3)$$

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Core: introductory factoring

13. Factor completely; check by multiplying.

$$x^2 + 7x + 10$$

14. Factor completely; check by multiplying.

$$x^2 - 7x + 12$$

15. Factor completely; check by multiplying.

$$x^2 + 4x - 12$$

16. Factor completely; check by multiplying.

$$x^2 - 4x - 21$$

17. Factor completely; check by multiplying.

$$6x^2 + 12x$$

18. Factor completely; check by multiplying.

$$x^2 - 16$$

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Extension: mixed polynomial practice

19. Multiply and simplify.

$$(x^2 + 2x + 1)(x^2 + x - 2)$$

20. Multiply and simplify.

$$(2x^2 + x - 3)(x^2 - x + 2)$$

21. Multiply and simplify.

$$(x + 2)(x^2 - 4)$$

22. Factor completely and verify.

$$x^2 - 3x - 28$$

23. Factor completely and verify.

$$x^2 - 10x + 25$$

24. Explain the error: only the first and last terms were multiplied.

$$(x + 2)(x + 3) = x^2 + 6$$